

EECS 841: Computer Vision

Lecture 5

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Outline (Chapter 3)

Lecture 5_ Point Transform _Part II

Image Transformation: Different ways to transform an image into another image

- Geometric transform
- Point transform

Outline

Different ways to transform an image into another image

- Geometric transform
- Point transform (histogram)

GRAYLEVEL HISTOGRAMS

Graylevel Histograms

- **Histogram:** a simple but powerful technique for capturing the statistics of any type of data.
 - The space in which the data reside is divided into bins, and the histogram records the number of occurrences in each bin.
- **Graylevel histogram:** a histogram of image gray levels
- **Normalized histogram:** computed from the histogram by simply dividing each value by the total number of pixels in the image
- **Probability density function (PDF):** the normalized histogram

Question:

1. What is the highest value possible at PDF?
2. What is the lowest value possible at PDF?

Try this image:

4	3	4
4	2	3
5	5	7

Practice Problem: What is the histogram

Find the Histogram

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

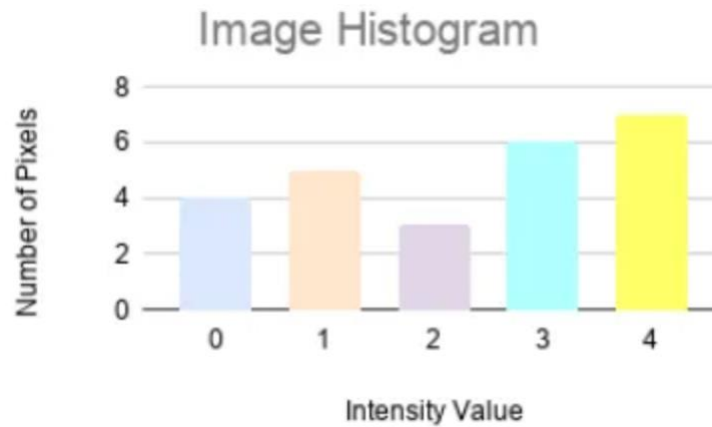


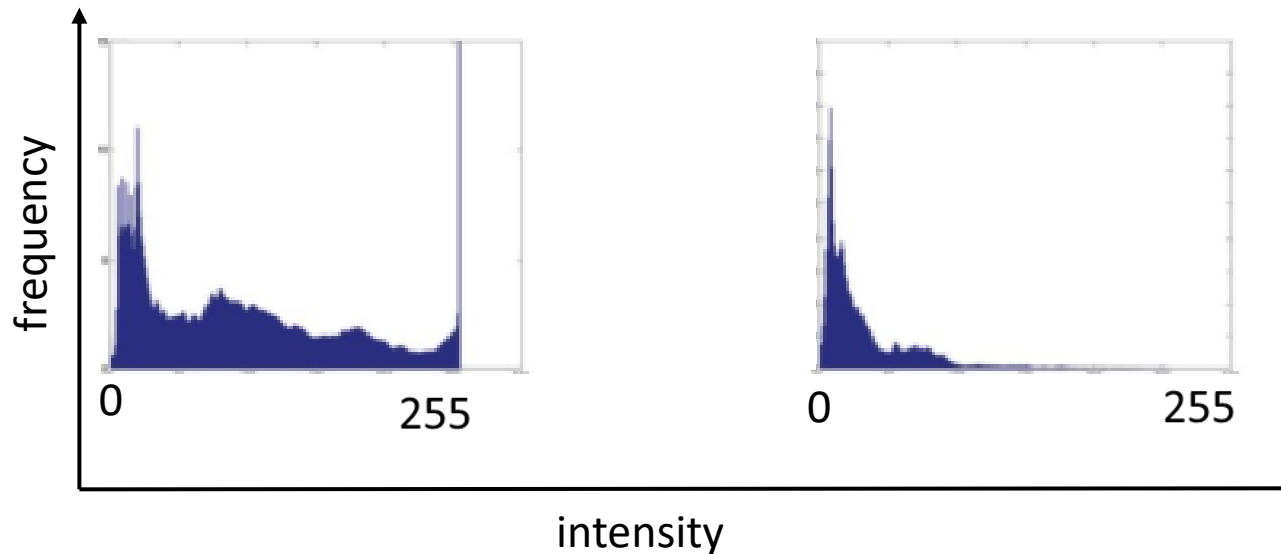
Image Histogram

Question: What information is interpreted from the histogram about the image?

Question: Looking at a histogram, you can guess about the overall intensity of the image. For a given histogram only (with no image), what information is missing about the image?

Graylevel Histograms

Figure 3.20 Images and their histograms. From left to right: an image with high contrast and many dark or bright pixels, a dark image with low contrast, and another high contrast image with good exposure. Note the spikes at 255 in the first and last images, indicating pixel saturation.

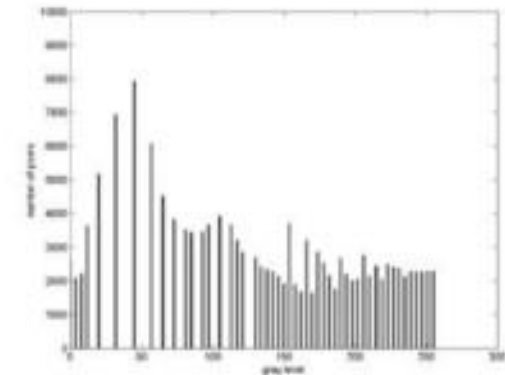
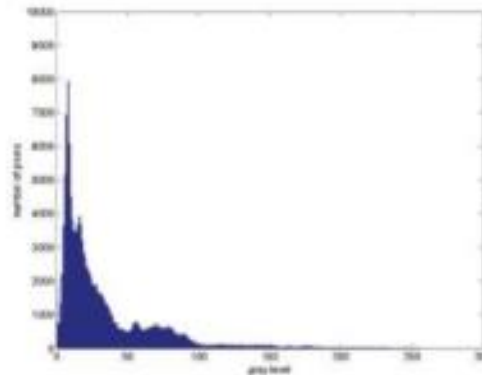


Histogram Equalization

Histogram equalization is the process of transforming the intensity values in an input image so that the histogram of the output image approximately matches a specified histogram to ensure a better contrast.

Histogram Equalization

Figure 3.21 The result of histogram equalization applied to an image. The increase in contrast is noticeable. The normalized histogram of the result is much flatter than the original histogram, but it is not completely flat due to discretization effects. Source: Movie *Hoop Dreams*.



How to Perform Histogram Equalization

1. Determine two parameters from the input image: maximum intensity level (K_{max}), and individual levels
 - Maximum intensity level = K_{max} = highest intensity (either you pick it from the input image or also can choose one nearby to the present current value)
 - Individual levels = 0, 1, 2, ... , K_{max} (starting from 0, write down all the individual levels up to K_{max})

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	$K_{max} * CDF$ (= y)	Floor (y)

How to Perform Histogram Equalization

2. Complete column 1 with the information from step 1 and move forward to count frequency

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
4 (=Kmax)					

How to Perform Histogram Equalization

2. Complete column 1 with the information from step 1 and move forward to count frequency

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0					
1					
2					
3					
4 (=Kmax)					

How to Perform Histogram Equalization

3. Count frequency

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

Intensity	Number of Pixels
0	4
1	5
2	3
3	6
4	7

Histogram Table

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4				
1	5				
2	3				
3	6				
4 (=Kmax)	7				

How to Perform Histogram Equalization

4. Determine PDF

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16			
1	5	.2			
2	3	.12			
3	6	0.24			
4 (=Kmax)	7	0.28			

How to Perform Histogram Equalization

5. Determine CDF

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16		
1	5	.2	.36		
2	3	.12	.48		
3	6	0.24	0.72		
4 (=Kmax)	7	0.28	1		

How to Perform Histogram Equalization

6. Determine $K_{max} * CDF$

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	$K_{max} * CDF$ (= y)	Floor (y)
0	4	.16	.16	0.64	
1	5	.2	.36	1.44	
2	3	.12	.48	1.92	
3	6	0.24	0.72	2.88	
4 (=Kmax)	7	0.28	1	4	

How to Perform Histogram Equalization

7. Take floor of y

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16	0.64	0
1	5	.2	.36	1.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4 (=Kmax)	7	0.28	1	4	4

How to Perform Histogram Equalization

Ans so far:

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

Intensity	Number of Pixels
0	4
1	5
2	3
3	6
4	7

Histogram Table

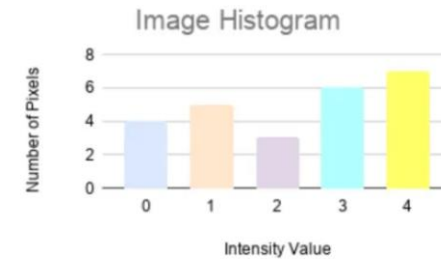


Image Histogram

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16	0.64	0
1	5	.2	.36	1.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4 (=Kmax)	7	0.28	1	4	4

Total pixel =25

- Answer 1: New histogram for equalized image

How many parameters do we need to know to find the new histogram?

Ans: 2

What are those?

Ans: new intensity vs new frequency

How to Perform Histogram Equalization

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16	0.64	0
1	5	.2	.36	0.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4 (=Kmax)	7	0.28	1	4	4

Total pixel =25

New intensity values (= new graylevels) in the equalized image	0	1 (1 appeared twice, but we will report just once)	2	4
New Frequency of the pixel is equalized image	4	8 (=5+3)	6	7

Total pixel = still 25!

Ans1: New Histogram

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16	0.64	0
1	5	.2	.36	0.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4 (=Kmax)	7	0.28	1	4	4

Total pixel =25

New intensity values (= new graylevels) in the equalized image	0	1 (1 appeared twice, but we will report just once)	2	4
New Frequency of the pixel is equalized image	4	8 (=5+3)	6	7

Total pixel = still 25!

Can you find the new histogram?

Ans1: New Histogram

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

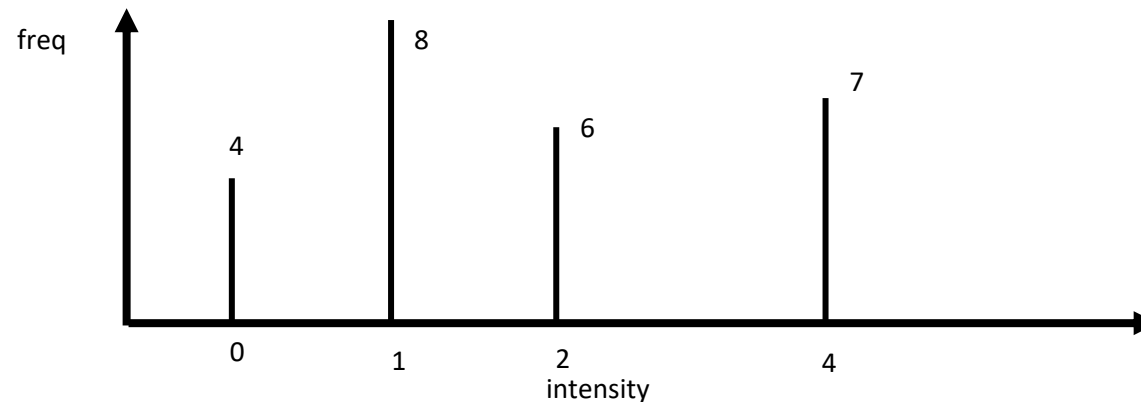
Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y)
0	4	.16	.16	0.64	0
1	5	.2	.36	0.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4 (=Kmax)	7	0.28	1	4	4

Total pixel =25

New intensity values (= new graylevels) in the equalized image	0	1 (1 appeared twice, but we will report just once)	2	4
New Frequency of the pixel is equalized image	4	8 (=5+3)	6	7

Total pixel = still 25!



- Answer 2: Find the new equalized image

How to Do Histogram Equalization

How to update image based on Histogram equation: mapping!

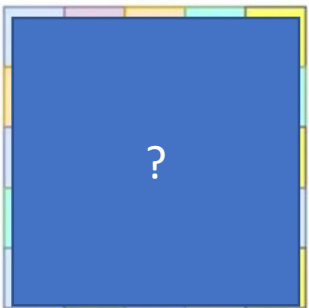
0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y) = suggested pixel intensity in equalized image
0	4	.16	.16	0.64	0
1	5	.2	.36	0.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4	7	0.28	1	4	4

= intensity in raw
image

= suggested intensity in the equalized
image



Modified: Image A

Previous 4 pixels with intensity of "0" will remain same (as the right most column also says so)

Previous 5 pixels with intensity "1" will remain "1" for the same

Previous 3 pixels with intensity "2" will be updated as "1" as suggested by right most column

Previous 6 pixels with intensity "3" will be updated as "2" as suggested by right most column

Previous 7 pixels with intensity "4" will remain same (as the right most column also says so)

How to Do Histogram Equalization

How to update image based on Histogram equation: mapping!

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

Image

# of Graylevel	Frequency (=n)	PDF (= n/ total pixels)	CDF	Kmax*CDF (= y)	Floor (y) = suggested pixel intensity in equalized image
0	4	.16	.16	0.64	0
1	5	.2	.36	0.44	1
2	3	.12	.48	1.92	1
3	6	0.24	0.72	2.88	2
4	7	0.28	1	4	4

0	1	1	2	4
1	2	4	2	2
0	1	2	1	4
2	1	4	1	0
0	4	1	4	4

Modified: Image A

Previous 4 pixels with intensity of "0" will remain same (as the right most column also says so)

Previous 5 pixels with intensity "1" will remain "1" for the same

Previous 3 pixels with intensity "2" will be updated as "1" as suggested by right most column

Previous 6 pixels with intensity "3" will be updated as "2" as suggested by right most column

Previous 7 pixels with intensity "4" will remain same (as the right most column also says so)